The same certificate bound holds for degree-two functions on perfect matchings

Gilad Chase, Neta Dafni, Yuval Filmus, Nathan Lindzey

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Editorial note. The counted statement is the author's labelled environment at source0.tex lines 490-492 and its proof environment opens at line 493; the pairing is by adjacency and the retained proof range is the whole environment.

Editorial note. Preamble: the author's own package lines, theorem-environment family and macro block (source lines 5-29).

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Editorial note. r002 note: r001 failed with an undefined reference raised from inside the counted proof; r002 adds exactly the environment that the citation names, as a context span with its provenance, and changes nothing else. The article's own compiled bibliography (main.bbl) was already bundled and is unrelated to this failure.

Editorial scope. One original result of the pinned author manuscript, counted once: the certificate bound for degree-two functions on perfect matchings (author label lem:C-bound-pms).

Allowed standard prerequisites (editorial).

- Degree of a Boolean function on matchings. The degree of f as a multilinear polynomial in the variables recording which pairs occur in the matching.. It is the hypothesis bounded by two.
- Certificate of a matching. A set of pairs prescribed to occur, whose size is the quantity bounded.. It is the quantity bounded by n minus two.
- Four-bit reduction. A certificate of size n minus one leaves just enough room to read off a four-bit function by choosing four independent alternatives in the matching.. It is the construction that produces the contradiction.

Lemma 0.1. *If $n \geq 8$ and $f\colon S_n \to \{0,1\}$ has degree at most 2, then $C(f) \leq n-2$.*

Lemma 0.2. *If $n \geq 8$ and $f\colon \mathcal{M}_{2n} \to \{0,1\}$ has degree at most 2, then $C(f) \leq n-2$.*

Proof. The proof is by contradiction. Suppose that $n \geq 8$, that $f\colon \mathcal{M}_{2n} \to \{0,1\}$ has degree at most 2, and that $C(f) = n-1$, say $C(f,m) = n-1$ for $m = \big\{\{1,2\},\{3,4\},\ldots,\{2n-1,2n\}\big\}$. Since $C(f,m) > n-2$, $m \setminus \big\{\{2i-1,2i\},\{2j-1,2j\}\big\}$ is not a certificate for m for any $i \neq j \in [n]$, and so $f(m^{ij}) \neq f(m)$, where m^{ij} is obtained from m by replacing $\big\{\{2i-1,2i\},\{2j-1,2j\}\big\}$ with $\big\{\{2i-1,2j\},\{2j-1,2i\}\big\}$.

We construct a function $g\colon \{0,1\}^4 \to \{0,1\}$ as follows: $g(y_1,y_2,y_3,y_4)$ is the value of f on the perfect matching obtained from m by applying the subset of the operations ^{12},^{34},^{56},^{78} specified by the input (for example, if $y_1 = 1$ then we apply ^{12}). As in the proof of Theorem 0.1, this results in a function of degree at most 2 satisfying $g(0,0,0,0) \neq g(1,0,0,0) = g(0,1,0,0) = g(0,0,1,0) = g(0,0,0,1)$, which is impossible. □

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